

electricity - practice problems

name _____

1. A small sensor wire in a greenhouse carries data from a humidity probe that checks the air every 10 seconds. If 3.6 C of charge passes through the wire in 2.0 minutes, what is the current?
2. When you press a doorbell button, 0.0060 C of charge passes one point in the circuit in 25 ms. What is the current during that pulse?
3. A light turns on almost instantly when a switch is closed, but electrons drift through the wire slowly. How can both statements be true?
4. A phone charges at 2.0 A for 1.5 hours. How much charge flows into the phone? Phone batteries are often labeled in mAh. How does your answer compare to a 3000 mAh battery?
5. A small motor spins a plastic fan with 7 blades. During one test, 18 C of charge passes through the motor while the current is 0.75 A. How long does the motor run?

6. About 2.5×10^{15} electrons drift to the right past one point in a wire in 0.20 s. What are the size and direction of the conventional current?
7. An ammeter reads 0.30 A in the wire leading into a light bulb. What would it read in the wire coming out of the bulb?
8. A battery is connected to a wire, but the wire is broken at one point. Can there still be voltage across the break? Can there be steady current around the circuit?
9. A 9.0 V battery is connected across a $330 \, \Omega$ resistor on a breadboard. What current flows through the resistor?
10. A small lamp uses 20 mA from a 1.5 V battery. What is the lamp's resistance?
11. A current of 18 mA flows through a $470 \, \Omega$ resistor. What voltage is across the resistor?

12. An indicator light needs 3.0 mA when it has 9.0 V across it. What resistance would give that current?

13. A $220\ \Omega$ resistor is connected to a battery for 5.0 s. What current flows through it?

14. A student measures the current through a resistor at three different voltages. Is the resistor ohmic (does it follow Ohm's law)? If so, what is its resistance?

voltage (V)	2.0	4.0	6.0
current (mA)	10	20	30

15. A 9.0 V battery pushes 3.0 C of charge through a circuit. How much energy does the battery give that charge?

16. A resistor stays the same. What happens to the current if the voltage across it doubles? What happens if the voltage stays the same but the resistance doubles?

17. A portable radio's 12 V battery pack is tested by connecting it across a $680\ \Omega$ resistor for 3.0 minutes. How much charge passes through the resistor?
18. A $2.2\ \text{k}\Omega$ resistor has a current of 1.2 mA. What voltage is across it, and how much charge passes in 45 s?
19. A toy motor draws 0.60 A from a 3.0 V battery while lifting a small paper flag. What is its effective resistance while running?
20. A 24 V source is connected across a resistor. In 15 s, 1.8 C of charge passes through the resistor. What is the resistance?
21. A resistor should allow 0.050 C to pass in 2.0 s when connected to a 6.0 V source. What resistance is needed?